

1

The main characteristics of a Darlington Amplifier are_____.

- ☐ a. Low input impedance, low output impedance and low voltage gain.
- ☐ b. High input impedance, high output impedance and high current gain.
- ☐ c. Low input impedance, low output impedance and high current gain.
- ☐ d. High input impedance, low output impedance and high current gain.

Question 2

Not yet answered

Marked out of 1.00

Flag question

What configuration is widely preferred in cascading amplifiers?

- ☐ a. Common Collector
- ☐ b. Common Base
- ☐ c. Common source
- ☐ d. Common Emitter

Question 3

Not yet answered

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Flag question

Enhancement mode MOSFETs are more commonly used as _____.

- ☐ a. Capacitors
- ☐ b. Buffers
- ☐ c. Resistors
- ☐ d. Switches

Question 4

Not yet answered

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Flag question

Which one of the following flow chart symbol is used for decision

- ☐ a. 
- ☐ b. 
- ☐ c. 
- ☐ d. 

on 5

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out of
question

The simplified form of Boolean expression $(X+Y+XY)(X+Z)$ is:

- ☐ a. $XZ+Y$
- ☐ b. $XY+YZ$
- ☐ c. $\bar{X}+Y+Z$
- ☐ d. $X+YZ$

6

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question

The forward characteristic of a diode has a slope of approximately 20mA/V at a desired point. The approximate incremental resistance of the diode is:

- ☐ a. 10 Ω
- ☐ b. 35 Ω
- ☐ c. 50 Ω
- ☐ d. 20 Ω

Time left 2:05

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g question

An n-channel D-MOSFET with a positive V_{GS} is operating in_____.

- ☐ a. the enhancement mode
- ☐ b. Saturation
- ☐ c. cutoff
- ☐ d. the depletion mode

n 8

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out of
question

The process of emitting photons from a semi-conductive material is called_____.

- ☐ a. electroluminescence.
- ☐ b. gallium arsenide.
- ☐ c. gallium phosphide.
- ☐ d. photoluminescence

on 9

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question

When the temperature of a doped semiconductor is increased, its conductivity;

- ☐ a. Increases.
- ☐ b. Does not change
- ☐ c. Increases or decreases depending on whether it is p- or n-type
- ☐ d. Decreases.

10

ut of

question

A signal $x(t)$ and its Hilbert transform is having _____

- ☐ a. Same energy spectral density
- ☐ b. Same amplitude spectrum
- ☐ c. Same power spectral density
- ☐ d. Same autocorrelation function

n 11

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out of

question

Which of the following type of circuits in electrical engineering cannot be analyzed using Ohm's law?

- ☐ a. Unilateral circuit
- ☐ b. Conductor circuit
- ☐ c. Bilateral circuit
- ☐ d. Linear circuit

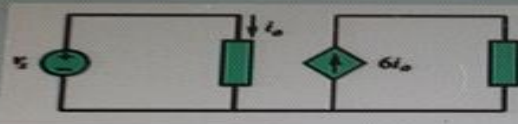
n 12

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question

The dependent source in figure below is:



- ☐ a. current-controlled current source
- ☐ b. current-controlled voltage source
- ☐ c. voltage-controlled voltage source
- ☐ d. voltage-controlled current source

Question 13

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The 1's complement of $(10111001)_2$ is

- ☐ a. $(01000110)_2$
- ☐ b. $(11000110)_2$
- ☐ c. $(01000111)_2$
- ☐ d. $(1010101)_2$

Question 14

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out of
question

Which of the following is true about open circuit?

- ☐ a. Circuit element with resistance approaching zero.
- ☐ b. Circuit element with current approaching infinity.
- ☐ c. Circuit element with voltage approaching zero.
- ☐ d. Circuit element with resistance approaching infinity.

Question 15

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question

Which of the following techniques is used in determining the time domain response of a linear time invariant system?

- ☐ a. Fourier Series
- ☐ b. Laplace transform
- ☐ c. Convolution Integral
- ☐ d. Fourier transform

Question 16

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lag question

In a PNP transistor, the current carriers are _____.

- ☐ a. acceptor ions
- ☐ b. holes
- ☐ c. free electrons
- ☐ d. donor ions

ion 17

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g question

The data elements in structure are also known as:

- ☐ a. Objects
- ☐ b. Instruction
- ☐ c. Members
- ☐ d. Data

18

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question

Which among the following assertions represents a necessary condition for the existence of Fourier Transform of discrete time signal (DTFT)?

- ☐ a. Discrete Time Signal should be absolutely integrable
- ☐ b. Discrete Time Signal should be absolutely multipliable
- ☐ c. Discrete Time Signal should be absolutely summable
- ☐ d. Discrete Time Signal should be absolutely differentiable

Next ques

n 19

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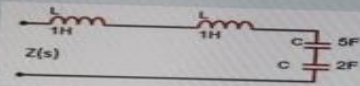
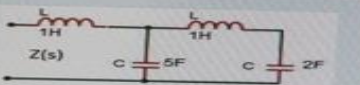
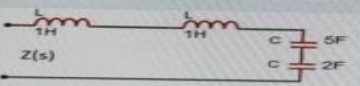
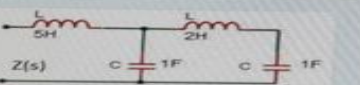
A type of assembly language instruction that alters the course of the program is called a -

- ☐ a. flow chart
- ☐ b. loop and jump
- ☐ c. loop
- ☐ d. jump

n 20

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out of
question

Realize the network in Cauer form $Z(s) = \frac{10s^4 + 12s^2 + 1}{2s^2 + 2s}$

- ☐ a. 
- ☐ b. 
- ☐ c. 
- ☐ d. 

According to Rayleigh's theorem, it becomes possible to determine the energy of a signal by:

- ☐ a. Estimating the area under the one-fourth power of its amplitude spectrum.
- ☐ b. Estimating the area exactly half as that of its amplitude spectrum
- ☐ c. Estimating the area under the square root of its amplitude spectrum.
- ☐ d. Estimating the area under the square of its amplitude spectrum.

Which one of the following is an example of a hardware:

- ☐ a. Web editor
- ☐ b. Instruction
- ☐ c. Memory
- ☐ d. Program

How many full adders are required to build a 7-bit ripple carry adder?

- ☐ a. 7
- ☐ b. 2^6
- ☐ c. 8
- ☐ d. 2^7

The invalid state of an S-R latch occurs when

- ☐ a. $S=1, R=0$
- ☐ b. $S=0, R=1$
- ☐ c. $S=0, R=0$
- ☐ d. $S=1, R=1$

Question 25

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g question

In which of the following header file is cin and cout defined in C++?

- ☐ a. < iomanip. h >
- ☐ b. < iostream. h >
- ☐ c. < fstream. h >
- ☐ d. < iocode. h >

Question 26

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out of

question

In an N-type semiconductor, the position of Fermi-level:

- ☐ a. In conduction Band
- ☐ b. In valance Band
- ☐ c. Near valance Band
- ☐ d. Near Conduction Band

Question 27

out of

question

Time left 2:01

A pulse is applied to each input of an Exclusive-OR gate. One pulse goes HIGH at $t=0$ and goes back LOW at $t=1\text{ms}$. The other pulse goes HIGH at $t=0.8\text{ms}$ and goes back LOW at $t=3\text{ms}$. The output pulse can be described as follows:

- ☐ a. It goes HIGH at $t=0$ and back LOW at $t=3\text{ms}$
- ☐ b. It goes HIGH at $t=0$ and back LOW at $t=0.8\text{ms}$ then back to HIGH at $t=1\text{ms}$ and back low at $t=3\text{ms}$
- ☐ c. It goes HIGH at $t=0$ and back LOW at $t=0.8\text{ms}$
- ☐ d. It goes HIGH at $t=1\text{ms}$ and back LOW at $t=3\text{ms}$

Question 28

out of

question

For a two-port network, the condition of symmetry in terms of Y-parameters

- ☐ a. $h_{11} = -h_{22}$
- ☐ b. $Y_{11} = Y_{22}$
- ☐ c. $Y_{12} = Y_{21}$
- ☐ d. $Y_{12} = -Y_{21}$

on 29

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question

The output of an OR gate is LOW when:

- ☐ a. Any input is HIGH
- ☐ b. All inputs are LOW
- ☐ c. All inputs are HIGH
- ☐ d. one input is LOW

n 30

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out of
question

The total gain of a multistage amplifier is less than the product of the gains of individual stages due to _____.

- ☐ a. The use of many capacitors
- ☐ b. Power loss in the coupling device
- ☐ c. The use of many transistors
- ☐ d. Loading effect of the next stage

on 31

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out of
question

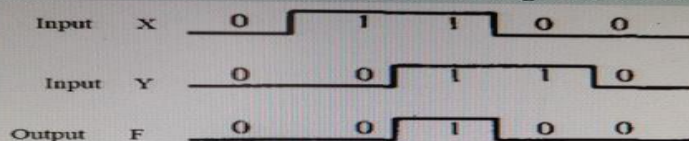
Int a=5; the word int shows

- ☐ a. Type
- ☐ b. Value
- ☐ c. Variable
- ☐ d. Operator

32

ut of
question

This item is based on the following electrical circuit



What is the type of the logic gate represented by the diagram?

- ☐ a. OR Logic gate
- ☐ b. AND Logic gate
- ☐ c. NOR Logic gate
- ☐ d. NAND Logic gate

n 33

What function does booting perform when the computer is switched on?

- ☐ a. Shut-down test
- ☐ b. Reliability test
- ☐ c. Power-on-self test
- ☐ d. Correct functioning test

n 34

To measure a current out of the second resistor in a series circuit consisting of four resistors, an Ammeter can be placed

- ☐ a. At the negative terminal of the source.
- ☐ b. Between the third and the fourth resistor.
- ☐ c. Between the second and the third resistor.
- ☐ d. At any point in the circuit.

35

A two-port network is defined by the relation: $I_1 = 5 V_1 + 3 V_2$
 $I_2 = 2 V_1 - 7 V_2$

The value of z_{22} is

- ☐ a. 7Ω
- ☐ b. $2/31 \Omega$
- ☐ c. -7Ω
- ☐ d. $-5/41 \Omega$

n 36

The domain of the expression $A\bar{B}CD + A\bar{B} + \bar{C}D + B$ is:

- ☐ a. A and D
- ☐ b. A, B, C, and D
- ☐ c. B only
- ☐ d. C only

37

Convert $(1011.101)_2$ binary number to decimal number

- ☐ a. $(11.625)_{10}$
- ☐ b. $(41.5)_{10}$
- ☐ c. $(11.5)_{10}$
- ☐ d. $(41.625)_{10}$

38

Which of the following active filters experiences propagation delay at the input type frequency spectrum?

- ☐ a. Butterworth filter
- ☐ b. Elliptical filter
- ☐ c. Chebyshev filter
- ☐ d. Bessel filter

39

What will be displayed by the following code?

```
int i = 0;
while(i <= 5) {
    i++;
}
cout << i << endl;
```

- ☐ a. 0 1 2 3 4 5
- ☐ b. 6
- ☐ c. 5
- ☐ d. 12345

40

An example of a peripheral unit is

- ☐ a. The video monitor
- ☐ b. The MPU
- ☐ c. The interphase adapter
- ☐ d. The address register

41

If three bulbs are connected in parallel and two bulb blows out, what happens to the other bulb?

- ☐ a. The other bulb blows out as well
- ☐ b. The other bulb stops glowing
- ☐ c. The other bulb glows with increased brightness
- ☐ d. The other bulb continues to glow with the same brightness

42

When the crystal diode current is large, the bias is _____.

- ☐ a. reverse
- ☐ b. forward
- ☐ c. small
- ☐ d. large

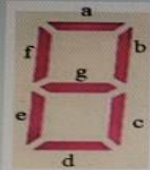
43

What happens to bandwidth if the total gain increased in cascade amplifiers?

- ☐ a. Increases
- ☐ b. Infinite
- ☐ c. Decreases
- ☐ d. Remain the same

44

This item is based on the diagram of 7-segment display shown below



A BCD to 7-segment display decoder has 0100 on its inputs. The active outputs are

- ☐ a. *b, c, f, g*
- ☐ b. *b, d, e, g*
- ☐ c. *a, c, f, g*
- ☐ d. *b, c, e, f*

45

The ratio of the voltage transform at the first port to the voltage transform at the second port is known as?

- ☐ a. Transfer admittance
- ☐ b. Current transfer ratio
- ☐ c. Voltage transfer ratio
- ☐ d. Transfer impedance

46

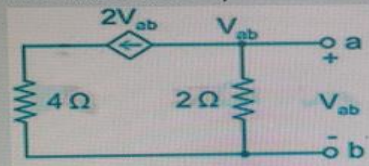
Current flowing through an element is $i = \begin{cases} 2t & 0 < t < 1 \\ 2t^2 & t > 1 \end{cases}$

The value of charge entering the element from $t = 0$ to $t = 2$ s.

- ☐ a. 0.333C
- ☐ b. 6.667C
- ☐ c. 6.333C
- ☐ d. 3.333C

47

The Thevenin equivalent resistance (R_{Th}) for the circuit shown in figure below is



- ☐ a. 2Ω
- ☐ b. 0.5Ω
- ☐ c. 0.4Ω
- ☐ d. 4Ω

48

Which of the following is correct about the voltage transformation ratio in electrical engineering?

- ☐ a. Ratio of secondary current to the primary current.
- ☐ b. Ratio of induced emf in secondary to induced emf in primary.
- ☐ c. Ratio of power in primary to power in secondary.
- ☐ d. Ratio of number of primary turns to the number of secondary turns.

49

The system function of a network has a response $R(s) = \frac{3}{(s^3 + 5s^2 + 6s)}$ and the excitation $E(s) = \frac{3}{s}$ then, Find $H(s)$

- ☐ a. $\frac{(s^4 + 5s^3 + 6s^2)}{9}$
- ☐ b. $\frac{s}{(s^2 + 5s + 6)}$
- ☐ c. $\frac{1}{(s^2 + 5s + 6)}$
- ☐ d. $\frac{9}{(s^4 + 5s^3 + 6s^2)}$

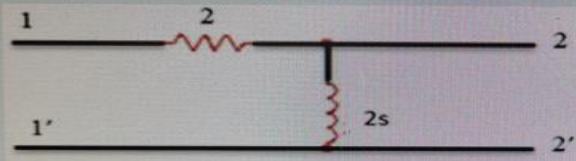
50

The lowest output impedance is obtained in case of BJT amplifiers for;

- ☐ a. CB configuration.
- ☐ b. CC configuration.
- ☐ c. CE configuration.
- ☐ d. CE with RE configuration.

51

Determine the driving point impedance $Z_{21}(s)$ for the network shown below



- ☐ a. $2(s + 2)$
- ☐ b. $s + 2$
- ☐ c. $s + 1$
- ☐ d. $2(s + 1)$

52

You have an application for a diode to be used in a tuning circuit. A type of diode to use might be;

- ☐ a. a Gunn diode.
- ☐ b. a Schottky diode.
- ☐ c. a varactor diode.
- ☐ d. an LED.

53

Thevenin's resistance R_{th} is found

- ☐ a. Between any two open terminals.
- ☐ b. By removing voltage sources along with their internal resistances.
- ☐ c. By short-circuiting the given two terminals.
- ☐ d. Between same open terminals as for E_{th}

54

When analyzing two port networks in parallel, which of the following is more convenient to use?

- ☐ a. z-parameters
- ☐ b. y-parameters
- ☐ c. h-parameters
- ☐ d. T-parameters

55

Which of the following is the correct syntax of including a user defined header files in C++?

- ☐ a. #include
- ☐ b. #include [userdefined]
- ☐ c. #include
- ☐ d. #include "userdefined"

56

Tomas is willing to do his final thesis project on high gain and high tuning filter. Depending on his idea; select the appropriate filter for his project title.

- ☐ a. Passive filter
- ☐ b. RL filter
- ☐ c. Active filter
- ☐ d. RC filter

57

A Laplace Transform exists when ____ i. The function is piece-wise continuous ii. The function is of exponential order iii. the function is piecewise discrete iv. The function is of differential order

- ☐ a. iii & iv
- ☐ b. i & iv
- ☐ c. i & ii
- ☐ d. ii & iii

58

A time domain signal is considered digital if,

- ☐ a. If it is has countable amplitude levels
- ☐ b. It is continuous along the time axis
- ☐ c. If it is discrete along the time axis
- ☐ d. If it has uncountable amplitude levels

59

Often a common-collector will be the last stage before the load. The main function(s) of this stage is;

- ☐ a. To provide voltage gain at the load
- ☐ b. To provide phase inversion of the input signal
- ☐ c. To buffer the voltage amplifiers from the low-resistance load and provide impedance matching for maximum power transfer
- ☐ d. provide a high-frequency path to improve the frequency response

60

A cylindrical conductor of length L and uniform cross-section A has resistance R . Another conductor of length $2L$ and resistance R of the same material has an area of cross-section of?

- ☐ a. $A/2$
- ☐ b. $2A$
- ☐ c. $3A/2$
- ☐ d. $3A$

61

A full adder is characterized by

- ☐ a. Three inputs and one outputs
- ☐ b. Three inputs and two outputs
- ☐ c. Two inputs and three outputs
- ☐ d. Two inputs and two outputs

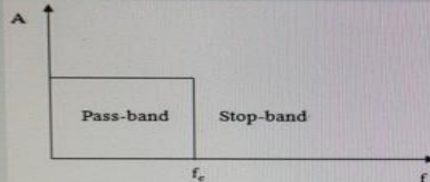
62

From the following options, which one is Hurwitz polynomial?

- ☐ a. $s^5 + 4s^4 + 7s^2 + 6s + 2$
- ☐ b. $s^4 + 4s^3 + 3s + 2$
- ☐ c. $s^6 + 7s^5 + 5s^4 - 3s^3 + 2s^2 + s + 4$
- ☐ d. $s^3 + 2s^2 + 4s + 2$

63

This item is based on the diagram of a filter response given below



In the above diagram of the filter response, what is the name of the filter response?

- ☐ a. Band stop filter response
- ☐ b. Low-pass filter response
- ☐ c. High pass filter response
- ☐ d. Band pass filter response

64

Determine the value of base x if $(211)_x = (152)_8$

- ☐ a. 4
- ☐ b. 8
- ☐ c. 7
- ☐ d. 9

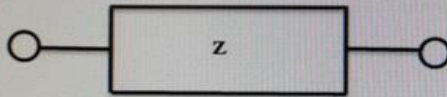
65

When the temperature of a doped semiconductor is increased, its conductivity ____.

- ☐ a. does not change
- ☐ b. increases
- ☐ c. depending on its p- or n-type
- ☐ d. decreases

66

This item is based on the following diagram of the symbol used in electrical network.



What is the name of this symbol in electrical network represented by letter 'Z' in the above diagram?

- ☐ a. Inductance
- ☐ b. Impedance
- ☐ c. Conductance
- ☐ d. Capacitance

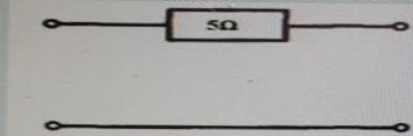
67

Which of the following according to KCL must be zero?

- ☐ a. Algebraic sum of currents entering and leaving a junction
- ☐ b. Algebraic sum of currents in closed-loop
- ☐ c. Algebraic sum of power in closed-loop
- ☐ d. Algebraic sum of voltages across the input and output

68

Find the $Y_{21}(s)$ parameter of the network shown below



- ☐ a. $-\frac{5}{1} \text{ mho}$
- ☐ b. $\frac{1}{5} \text{ mho}$
- ☐ c. $-\frac{1}{5} \text{ mho}$
- ☐ d. $\frac{5}{1} \text{ mho}$

69

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question

What is the output of the following program?

```
#include < iostream >
using namespace std;
int main() { int i, j, k;
i = 14;
j = 10;
k = (i > j) ? i : j;
cout << k << endl;
return 0;
}
```

- ☐ a. 20
- ☐ b. 14
- ☐ c. 10
- ☐ d. 24

70

d
out of
question

What do the negative frequency components represent after transforming a time domain signal to the frequency domain?

- ☐ a. The existence of a negative frequency
- ☐ b. Previously sampled frequency components
- ☐ c. The complex conjugate of the positive frequency components
- ☐ d. The phase shift in frequency components

71

d
out of
question

An LC circuit cannot produce oscillation, when

- ☐ a. Resistance is low
- ☐ b. Resistance is large
- ☐ c. Inductance is large
- ☐ d. Capacitance is large

72

d
out of
question

In which of the following header file is cin and cout defined in C++?

- ☐ a. < iostream. h >
- ☐ b. < iomanip. h >
- ☐ c. < fstream. h >
- ☐ d. < iocode. h >

73

What is largest number that can be represented using 6 bits?

- ☐ a. $(41.625)_{10}$
- ☐ b. $(41.5)_{10}$
- ☐ c. $(12)_{10}$
- ☐ d. $(11.625)_{10}$

74

Digital logic circuits in which the logic states of the inputs at any given instant determine the state of the output are called:

- ☐ a. Combinational logic circuits
- ☐ b. Sequential logic circuits
- ☐ c. Reasonable logic circuits
- ☐ d. Automated logic circuits

75

The signal is said to be _____ whenever we can define a signal at an instant of time

- ☐ a. Both deterministic and non-deterministic
- ☐ b. Deterministic
- ☐ c. Continues
- ☐ d. Non-Deterministic

76

The energy of power signal is equivalent to_____.

- ☐ a. Infinity
- ☐ b. One
- ☐ c. Very small
- ☐ d. Zero

n 77

d
out of
question

In C++ class declaration ends with which character?

- ☐ a. }
- ☐ b.)
- ☐ c. ;
- ☐ d. ::

n 78

d
out of
question

In superposition theorem, when we consider the effect of one voltage source, all the other voltage sources are;

- ☐ a. Removed
- ☐ b. Opened
- ☐ c. Undisturbed
- ☐ d. Shorted

n 79

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out of
question

A combinational circuit which is used to change a decimal number into an equivalent BCD number is:

- ☐ a. Decoder
- ☐ b. De-multiplexer
- ☐ c. Multiplexer
- ☐ d. Encoder

n 80

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out of
question

h- parameters are used in the analysis of:

- ☐ a. Open circuit analysis
- ☐ b. Electronic circuits
- ☐ c. Transmission line
- ☐ d. Short circuit analysis

on 81

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g question

Which one of the following is NOT true?

- ☐ a. Physical quantities such as position, velocity, acceleration & temperature are digital in nature.
- ☐ b. Analogue variables are digitized at the input with the help of an analog-to-digital converter
- ☐ c. Digital representation produces a discrete output
- ☐ d. Analogue representation gives a continuous output

on 82

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question

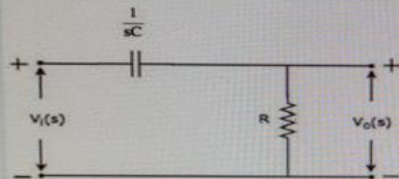
When transistors are used in digital circuits they usually operate in the ____.

- ☐ a. saturation and cutoff regions
- ☐ b. active region
- ☐ c. linear region
- ☐ d. breakdown region

n 83

d
out of
question

This item is based on the diagram of a filter network given below



In the above diagram of the filter network, what is the name of a filter network indicate?

- ☐ a. Low-pass filter
- ☐ b. Band stop filter
- ☐ c. Band pass filter
- ☐ d. High pass filter

84

i
out of
question

While Thevenizing a circuit between two terminals V_{th} is equal to

- ☐ a. E.M.F. of the battery nearest to the terminals
- ☐ b. Net voltage available in the circuit
- ☐ c. Short-circuit terminal voltage
- ☐ d. Open-circuit terminal voltage

85

out of

question

A filter that passes frequency between two designated cut-off frequencies f_c and attenuates all other frequencies greater than f_{cis}

- ☐ a. Band elimination filter
- ☐ b. High pass filter
- ☐ c. Low pass filter
- ☐ d. Band pass filter

86

out of

question

Which one of the following is the purpose of a wide band-pass filter?

- ☐ a. Zero noise
- ☐ b. Sharp cut-off frequency value
- ☐ c. Sharp bandwidth value
- ☐ d. Sharp bandwidth and sharp cut-off frequency value

87

ed

d out of

g question

Which of the following symbol is used to declare the preprocessor directives in C++?

- ☐ a. *
- ☐ b. \$
- ☐ c. ^
- ☐ d. #

88

If A_v , A_i and A_p represents the voltage gain, current gain and power gain ratio of an amplifier which of the following is correct expression for the corresponding values in decibel?

- ☐ a. Power gain: $(20 \log A_p) \text{ dB}$
- ☐ b. Power gain: $20 \log A_p \text{ dB}$
- ☐ c. Voltage gain: $(10 \log A_v) \text{ dB}$
- ☐ d. Current gain: $10 \log A_i \text{ dB}$

89

The Fourier series for the function $f(x) = \sin^2 x$ is _____

- ☐ a. $0.5 - 0.5 \cos 2x$
- ☐ b. $0.5 + 0.5 \cos 2x$
- ☐ c. $0.5 + 0.5 \sin 2x$
- ☐ d. $0.5 - 0.5 \sin 2x$

90

Which of the following is the brain of the computer?

- ☐ a. CPU
- ☐ b. Memory
- ☐ c. Control unit
- ☐ d. Arithmetic and Logic unit

